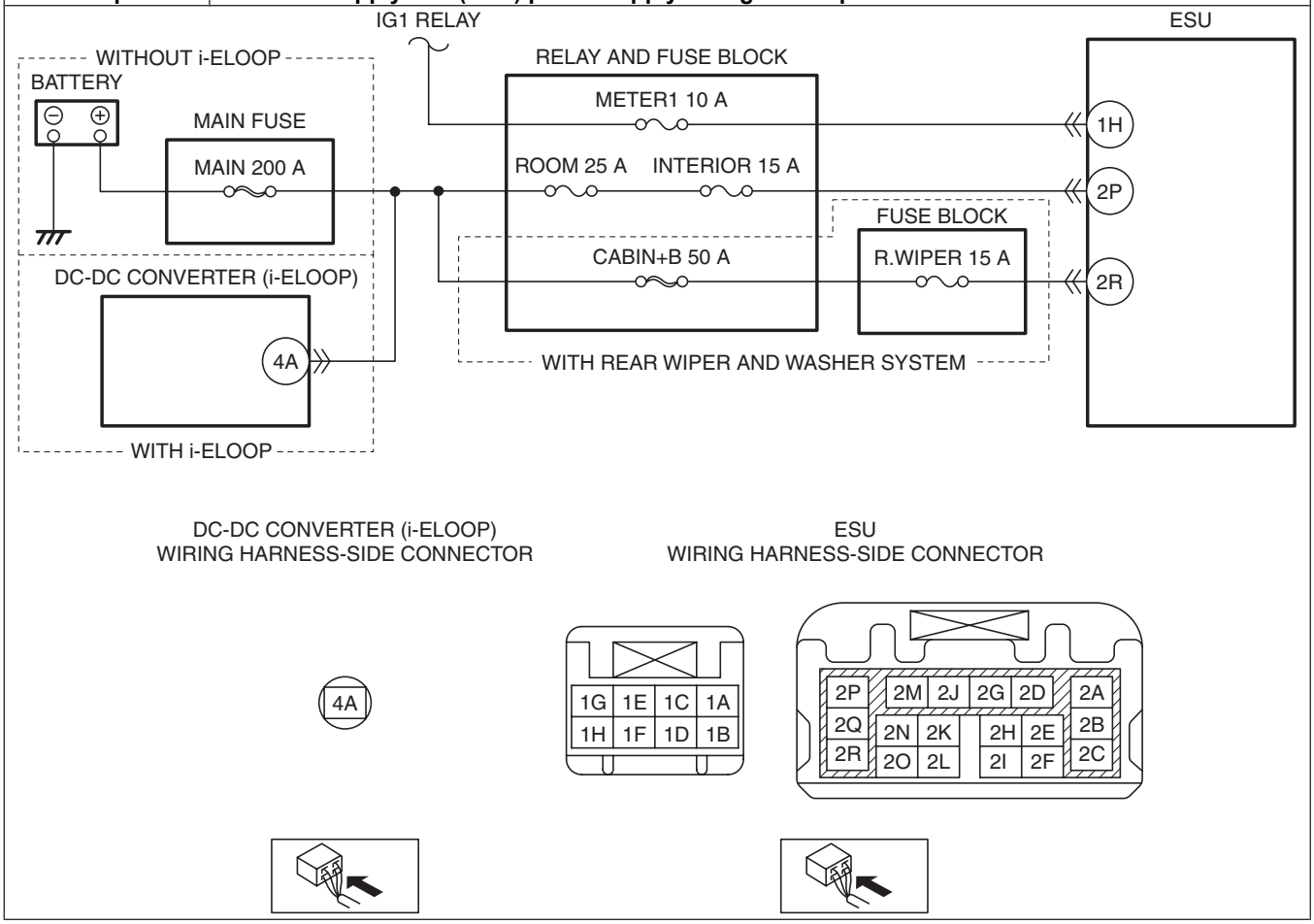
DTC U3003:16 [ELECTRICAL SUPPLY UNIT (ESU)]

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Description	Electrical supply unit (ESU) power supply voltage low input
Detection condition	• Electrical supply unit (ESU) power supply circuit (terminal 1H, 2P, or 2R) of **less than 8 V** is detected for **10 s or more**.
Fail-safe	• Not applicable
Possible cause	**Without i-ELOOP:** • DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Electrical supply unit (ESU) connector or terminal malfunction • Electrical supply unit (ESU) power supply circuit malfunction — Short to ground in the wiring harness between the following terminals: • METER1 10 A fuse—Electrical supply unit (ESU) terminal 1H • MAIN 200 A fuse—Electrical supply unit (ESU) terminal 2P • MAIN 200 A fuse—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) — METER1 10 A fuse malfunction — INTERIOR 15 A fuse malfunction — ROOM 25 A fuse malfunction — R.WIPER 15 A fuse malfunction (With rear wiper and washer system) — CABIN+B 50 A fuse malfunction (With rear wiper and washer system) — MAIN 200 A fuse malfunction — Open circuit in wiring harness between the following terminals: • IG1 relay—Electrical supply unit (ESU) terminal 1H • Battery positive terminal—Electrical supply unit (ESU) terminal 2P • Battery positive terminal—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) • Electrical supply unit (ESU) malfunction **With i-ELOOP:** • DTCs are stored in the PCM. • DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • Battery malfunction • Generator malfunction • Electrical supply unit (ESU) connector or terminal malfunction • Electrical supply unit (ESU) power supply circuit malfunction — Short to ground in the wiring harness between the following terminals: • METER1 10 A fuse—Electrical supply unit (ESU) terminal 1H • ROOM 25 A fuse—Electrical supply unit (ESU) terminal 2P • CABIN+B 50 A fuse—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) — METER1 10 A fuse malfunction — INTERIOR 15 A fuse malfunction — ROOM 25 A fuse malfunction — R.WIPER 15 A fuse malfunction (With rear wiper and washer system) — CABIN+B 50 A fuse malfunction (With rear wiper and washer system) — Open circuit in wiring harness between the following terminals: • IG1 relay—Electrical supply unit (ESU) terminal 1H • DC-DC converter (i-ELOOP) terminal 4A—Electrical supply unit (ESU) terminal 2P • DC-DC converter (i-ELOOP) terminal 4A—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) • Electrical supply unit (ESU) malfunction

Description	Electrical supply unit (ESU) power supply voltage low input



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	**Without i-ELOOP:** • Go to Step 3. **With i-ELOOP:** • Go to the next step.
2	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
3	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 8. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Step	Inspection		Action
4	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 8. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5	**INSPECT ELECTRICAL SUPPLY UNIT (ESU) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the electrical supply unit (ESU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	**Without i-ELOOP:** • Go to the next step. **With i-ELOOP:** • Go to Step 7.
		No	Repair or replace the connector, then go to Step 8.

Step	Inspection		Action
6	**VERIFY ELECTRICAL SUPPLY UNIT (ESU) POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the following terminals (wiring harness-side): — Electrical supply unit (ESU) terminal 1H — Electrical supply unit (ESU) terminal 2P — Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) • Is the voltage **B+**?	Yes	Go to Step 8.
		No	Inspect the METER1 10 A fuse, INTERIOR 15 A fuse, ROOM 25 A fuse, R.WIPER 15 A fuse (With rear wiper and washer system), CABIN+B 50 A fuse (With rear wiper and washer system) and MAIN 200 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • METER1 10 A fuse—Electrical supply unit (ESU) terminal 1H • MAIN 200 A fuse—Electrical supply unit (ESU) terminal 2P • MAIN 200 A fuse—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • IG1 relay—Electrical supply unit (ESU) terminal 1H • Battery positive terminal—Electrical supply unit (ESU) terminal 2P • Battery positive terminal—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.

Step	Inspection		Action
7	**VERIFY ELECTRICAL SUPPLY UNIT (ESU) POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the following terminals (wiring harness-side): — Electrical supply unit (ESU) terminal 1H — Electrical supply unit (ESU) terminal 2P — Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) • Is the voltage **B+**?	Yes	Go to the next step.
		No	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the METER1 10 A fuse, INTERIOR 15 A fuse, ROOM 25 A fuse, R.WIPER 15 A fuse (With rear wiper and washer system) and CABIN+B 50 A fuse (With rear wiper and washer system). • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • METER1 10 A fuse—Electrical supply unit (ESU) terminal 1H • ROOM 25 A fuse—Electrical supply unit (ESU) terminal 2P • CABIN+B 50 A fuse—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • IG1 relay—Electrical supply unit (ESU) terminal 1H • DC-DC converter (i-ELOOP) terminal 4A—Electrical supply unit (ESU) terminal 2P • DC-DC converter (i-ELOOP) terminal 4A—Electrical supply unit (ESU) terminal 2R (With rear wiper and washer system) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Connect the service plug, then go to the next step. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)

Step	Inspection		Action
8	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Switch the ignition ON (engine off or on) and wait for **10 s or more**. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the electrical supply unit (ESU). (See ELECTRICAL SUPPLY UNIT (ESU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
9	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].)
		No	DTC troubleshooting completed.